User Requirements Document

Project: Fragrance Dataset Overview Dashboard
Tools: MySQL, Tableau, GitHub, CSV/raw dataset

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1. Project Overview

This project required transforming a raw fragrance dataset into a cleaned, validated dataset and an accompanying Tableau dashboard. The raw data arrived in a messy state — duplicate records, inconsistent gender labels, text-based numeric fields, and list-like accord data — and needed to be processed in MySQL before any reliable analysis or visualization could be built.

2. Project Goal

The goal was to clean, deduplicate, validate, and visualize fragrance data in order to summarize trends by gender, brand, rating, and primary accords. The end result is a single-page Tableau dashboard backed by a quality-checked SQL view.

3. Target User

- **Portfolio reviewer / recruiter** – wants to quickly assess SQL, data cleaning, and visualization skills
- **Business stakeholder** – wants a high-level overview of fragrance dataset trends without touching the underlying data
- **Data analyst** – wants to review fragrance dataset trends and verify how the cleaned dataset was produced

4. Business Questions

The project needed to answer the following questions:

- How many fragrances are in the cleaned dataset?
- What is the average fragrance rating?
- What is the total rating count?
- How are fragrances distributed by gender?
- Which brands have the most fragrances?
- Which fragrances are highest rated with 100+ ratings?
- Which primary accords are most common?

5. Project Scope

In scope

- Import raw fragrance data into MySQL
- Remove duplicate fragrance records
- Clean and standardize key fields

- Run SQL quality checks
- Build a Tableau dashboard
- Document the process in GitHub

Out of scope

- Building a live production database
- Creating a real-time dashboard refresh pipeline
- Predicting fragrance ratings with machine learning
- Claiming the results represent the entire global fragrance market

6. Data Requirements

The project required raw fragrance data containing the following fields:

Field	Why it is needed
URL	Deduplication (via normalized URLs) and extracting perfume and brand information
Name	Fallback source for perfume names when URL extraction is not usable
Gender	Category analysis by gender (Men, Women, Unisex)
Rating Value	Rating analysis (average rating, top rated fragrances)
Rating Count	Rating analysis (total rating count, filtering for 100+ ratings)
Main Accords	Accord trend analysis (most common primary accords)

7. Functional Requirements

The solution needed to:

- Create a raw table (`fra_perfumes_raw`) for imported data
- Create a deduplicated table (`fra_perfumes_raw_deduped`) using normalized URLs
- Create a cleaned SQL view (`vw_fragrances_cleaned`)
- Extract perfume names from URLs, with the raw Name field as a fallback
- Extract brand names from URLs
- Standardize gender into Men, Women, and Unisex
- Convert rating values into numeric decimals
- Convert rating counts into numeric whole numbers
- Clean and split main accords into Main Accord 1 through Main Accord 5
- Connect Tableau to the cleaned dataset or exported cleaned view
- Build the required dashboard charts and KPIs

8. Data Quality Requirements

The cleaned dataset needed to satisfy the following rules before dashboard development:

- Duplicate URL check should return 0 rows
- Gender must only be Men, Women, or Unisex when classified
- Rating Value must be between 0 and 5
- Rating Count cannot be negative
- Cleaned perfume names should not become missing if raw name or URL data is usable
- Cleaned brand names should not become missing if the URL contains brand information
- Cleaned accord fields should not become missing if raw accord data exists

9. Dashboard Requirements

The dashboard must include:

- Total Fragrances KPI
- Average Rating KPI
- Total Rating Count KPI
- Fragrances by Gender chart
- Top Brands by Fragrance Count chart
- Top Rated Fragrances with 100+ Ratings chart
- Most Common Primary Accords treemap

10. Non-Functional Requirements

- Dashboard should be easy to read at a glance
- Charts should have clear titles and labels
- The dashboard should fit on one main overview page
- SQL should be organized into separate files by purpose (table creation, import, cleaning, quality checks)
- Screenshots should be saved as evidence of each project stage
- README should explain the final project clearly enough for someone to understand and reproduce it

11. Deliverables

Documentation

- `README.md`
- `user_requirements_document.md`
- `user_requirements_document.docx`

- `user_requirements_document.pdf`

SQL and Tableau files

- SQL files for table creation, import, cleaning, and quality checks
- Tableau workbook

Screenshot evidence

- Final dashboard screenshot
- Raw data preview screenshot
- Deduplicated table preview screenshot
- Cleaned view preview screenshot
- Row count check screenshot
- Duplicate URL check screenshot
- Missing values quality check screenshots (names, brands, accords)
- Validity check screenshots (gender, rating value, rating count)

12. Assumptions and Limitations

- Results depend on the provided raw dataset
- Null or unclear gender values may remain unclassified after standardization
- The dashboard describes this cleaned dataset and should not be interpreted as a complete ranking of the entire fragrance market
- Rating averages depend on records with valid numeric rating values; invalid or missing values are excluded

13. Acceptance Criteria / Success Criteria

The project is considered complete when all of the following are true:

- ☐ Raw data is imported successfully into MySQL
- ☐ Deduplicated table (`fra_perfumes_raw_deduped`) is created
- ☐ Cleaned SQL view (`vw_fragrances_cleaned`) is created
- ☐ Cleaned view returns 69,948 rows
- ☐ Duplicate URL check returns 0 rows
- ☐ Missing cleaned perfume name check returns 0 rows
- ☐ Missing cleaned brand check returns 0 rows
- ☐ Missing cleaned accords check returns 0 rows
- ☐ Invalid gender check returns 0 invalid values
- ☐ Invalid rating value check returns 0 invalid values
- ☐ Invalid rating count check returns 0 invalid values
- ☐ Tableau dashboard includes all required KPI cards and charts
- ☐ Project files and screenshots are organized in GitHub